# Adversarial Test-Hardening for AI-Written Code: An Instrument Autopsy and a Pre-Registered Causal Estimate of the Critic Loop

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**Abstract**

Large language models increasingly write both code and the tests meant to check it, and coverage, the usual measure of a suite, records what ran rather than what was verified. We study an adversarial test-hardening loop under a mechanical oracle: a Tester model writes tests, mutation testing names the surviving injected defects, and a Critic model writes tests to kill exactly those, with every verdict decided mechanically so that no model judges another model's output. We report two pre-registered experiments and the instrument autopsy that connects them. In Experiment 1, on five Python subjects (one same-lineage-loop cell could not be scored), the loop killed 105 mutants that one-shot generation missed and lost none, and the cross-lineage-Critic question (H2) returned a pre-declared null; the central finding was an autopsy: an earlier analysis had reported a cross-lineage effect at $p = 9.5 \times 10^{-66}$ that was a lineage-correlated instrument artifact, an output cap that silently truncated the verbose model, caught only by adversarial review of the completed analysis. A subsequent cross-model review of this manuscript identified a remaining confound: each arm resampled its own initial test suite, so Experiment 1's comparisons are between independently sampled pipelines. Experiment 2 removes it. Under a publicly pre-registered frozen-shared-round-0 design (five replicates on each of four subjects, every seed committed before its continuations), same-lineage Critic rounds killed 78% of the survivors the frozen initial suite left standing (mean incremental kill rate 0.783, 95% cluster-bootstrap interval [0.592, 0.935]), a within-replicate causal estimate; and the cross-provider critic configuration, which Experiment 1's design could not distinguish from its same-provider control, showed a positive pilot difference (within-replicate rate gap 0.178, 95% bootstrap interval [0.039, 0.347]; direction stable under every leave-one-out sensitivity, magnitude dominated by a single replicate) at 5.5x lower arm cost. This is a comparison of two named model-provider-harness configurations, not an isolated lineage effect: part of the measured gap is one configuration's receipted operational failures, including recurrences of the autopsy's truncation mechanism under the corrected instrument, this time detected and scored rather than laundered. We conclude that cross-model comparisons can inherit the asymmetries of the harness that runs them, that adversarial cross-model review catches what mechanical gates miss at the instrument, analysis, and manuscript levels alike, and we release both protocols, all receipts, and the analysis code.

Disclosure: This manuscript was drafted by an AI assistant from the author's frozen, pre-registered experimental artifacts under the author's direction; every quantitative and design claim was checked against the committed sources. The tooling is disclosed rather than hidden, consistent with the paper's subject (see Section 3.9).

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## 1. Introduction

Large language models now write both production code and the tests meant to check it, and the second job is the weaker link. Coverage, the usual measure of a test suite, records which lines ran, not whether anything about their behavior was actually verified, and a model asked to write tests for its own code tends to take the path of least resistance: it exercises the happy path and asserts weakly, producing suites that are green and nearly empty of fault-detecting power. A test suite can be fully covered and catch almost nothing. As models take over more of the testing work, the evidentiary value of "the tests pass" is arguably falling, precisely where practitioners are relying on it more.

Mutation testing is the standard way to measure a suite's fault-detecting power directly. It injects small, deliberate defects (mutants) into the code and counts how many the suite catches: a killed mutant is a defect the tests detected, and a surviving mutant is a specific, named gap the tests would let through. We use mutation testing as a mechanical oracle inside a two-role loop. A Tester model writes an initial suite; the surviving mutants are handed, by name, to a Critic model that writes tests to kill exactly those; the loop repeats until no new mutants die. Every verdict in this loop is mechanical: a generated test either kills a mutant under the test runner or it does not, and no model ever judges its own or another model's output. The design deliberately removes model judgment from the verdict, which is the channel through which self-preference bias corrupts model-graded evaluation. Removing model judgment from the verdict is necessary but, as this study's own experience shows (Section 5), not sufficient to make a cross-model comparison neutral.

We do not claim this loop is new. Feeding survivor mutants back into a model prompt and running an adversarial test-versus-mutant loop under a mechanical oracle are both established in prior work. Our study is a pre-registered empirical evaluation, and its value is in the discipline with which it was run and reported, not in the idea. We froze the hypotheses, the subjects, the measurement scopes, and the success criteria before any paid run, enforced that freeze in tooling, and required every reported number to be recomputable from committed per-run receipts. We test two hypotheses on five pinned Python subjects under the mutation-kill oracle: H1, that the adversarial loop kills more mutants than one-shot generation; and H2, that a cross-lineage Critic (a different-provider model) kills more of the survivors a same-lineage Tester missed than a same-lineage Critic does. We report cost per killed mutant throughout.

In Experiment 1, H1 is supported as a comparison of two independently sampled pipelines: across the paired mutants where the two approaches disagreed, the loop killed 105 that one-shot generation missed and lost none, a result that grew stronger, not weaker, when a measurement defect working against it was corrected. H2 is not supported there; it is a pre-declared null in a design that, as Experiment 2 shows, could not have distinguished the configurations. Because each Experiment 1 arm drew its own initial test suite, those comparisons do not isolate the loop or the critic lineage from initial-draw sampling variance, a limitation stated where the arms are defined (Section 3.2).

The paper's central contribution is neither of those results. An earlier version of the H2 analysis reported an overwhelming cross-lineage effect (pooled $p = 9.5 \times 10^{-66}$) that was almost entirely an instrument artifact: an undetected output ceiling silently truncating the verbose model's replies, correlated with exactly the variable under test. Section 4.4 reports that incident once, with receipts, and Section 5.2 interprets it once. The general lesson is that a cross-model comparison

can inherit the asymmetries of the harness that runs it: closing the model-judgment channel does not make the comparison neutral if the apparatus feeding the two models is not itself symmetric.

The confound was not caught by us. After Experiment 1 was analyzed and this manuscript drafted, a cross-model adversarial review of the manuscript itself identified the round-0 resampling confound as blocking any causal reading of H1. Rather than reframe the claim, we removed the confound: Experiment 2 was pre-registered and pushed publicly before its runner existed, freezing one Tester round-0 suite per replicate (five per subject, four subjects) and running all three continuations from that identical state, which makes the loop's incremental effect a within-replicate paired quantity. The same-lineage Critic rounds killed a mean 78% of the survivors the frozen round 0 left standing (incremental kill rate 0.783, 95% cluster-bootstrap interval [0.592, 0.935]); the cross-provider configuration showed a positive pilot difference (within-replicate rate gap 0.178 [0.039, 0.347]) at 5.5x lower arm cost; and the autopsy's mechanism recurred three times under the corrected instrument, this time detected, receipted, and scored honestly (Section 4.8).

This paper makes three contributions:

1. A causal, within-replicate estimate of the incremental effect of a guarded, add-only Critic continuation over a frozen initial suite, under a publicly pre-registered frozen-shared-round-0 design (Experiment 2), together with the descriptive pipeline-level comparison that preceded it (Experiment 1), with cost per killed mutant disclosed throughout, a normalization not found in the studies reviewed here.
2. A pilot comparison of a cross-provider against a same-provider Critic configuration under a mechanical mutation-kill oracle, a comparison we could not find in the studies reviewed here: indistinguishable under Experiment 1's independently-sampled design, a positive within-replicate difference under Experiment 2's paired design, reported with per-replicate values and leave-one-out sensitivities, and explicitly a configuration comparison (model, provider API behavior, and harness interaction bundled), never an isolated lineage effect.
3. As the durable contribution, a fully receipted autopsy of a provider-correlated instrument artifact, its fail-closed repair, and the mechanism's detected recurrences under the corrected instrument; together with the public artifact (both protocols, all receipts and frozen seeds, the deviations log, and the analysis scripts) and the outside-the-family review process through which each of these defects was caught.

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## 2. Related Work

### 2.1 LLM-based test generation

A large recent literature uses large language models to generate unit tests directly. TestPilot [arXiv:2302.06527, TSE 2024] generates JavaScript tests from signatures and documentation usage examples with no additional training; ChatTester [arXiv:2305.04207] and ChatUniTest [arXiv:2305.04764] add generate-then-repair loops that fix compilation and assertion errors in model-written tests. For Python specifically, CoverUp [arXiv:2403.16218, FSE 2025] interleaves coverage analysis with iterative model dialogs to target uncovered lines, and CodaMosa [ICSE 2023], building on the search-based generator Pynguin (Lukasczyk and Fraser), queries a model to re-seed search when coverage plateaus. HITS [arXiv:2408.11324, ASE 2024] decomposes complex

methods into slices for higher coverage. At industrial scale, TestGen-LLM [arXiv:2402.09171, FSE 2024] extends existing test classes only when a generated test verifiably builds, passes reliably, and increases coverage, using that filter to eliminate hallucinated tests. Adjacent lines of work target the oracle directly, generating assertions rather than whole suites (TOGA [arXiv:2109.09262, ICSE 2022]), or use models as fuzzers against crash and inconsistency oracles rather than regression suites (TitanFuzz [arXiv:2212.14834, ISSTA 2023]; Fuzz4All [arXiv:2308.04748, ICSE 2024]); we exclude fuzzing from the comparison because its oracle (crashes, differential behavior) and artifact (inputs, not persistent test suites) differ from ours. These systems establish that models can write useful tests and that a filter is needed to trust them, but they gate on coverage or on compile-and-pass repair, not on fault detection. Our work differs in the gate: it scores generated tests by the mutants they kill, a stronger signal than coverage, and it is the adversarial survivor-feedback loop, not one-shot generation, that our H1 evaluates. This study does not include a non-LLM search-based baseline such as Pynguin or EvoSuite; the comparison of interest here is one-shot versus adversarial-loop LLM generation, not LLM versus search-based generation.

### 2.2 Mutation testing with language models

Mutation testing supplies the mechanical oracle for this work. Two surveys frame what that oracle can and cannot claim. The oracle problem [Barr et al., IEEE TSE 2015] is the general difficulty of deciding what a test should assert; our pass-on-pristine validation plus mutant-kill scoring is a partial oracle, and the wrong-oracle tests we prune (generated tests that fail on correct code) are its concrete residue. Papadakis et al.'s mutation-testing survey [Advances in Computers 2019] treats equivalent mutants, mutation-score interpretation, and the non-independence of mutants, all three of which appear in this study as disclosed limitations rather than solved problems. Several systems combine mutation testing with language models. MuTAP [arXiv:2308.16557; IST 2024] re-prompts a single model with its surviving mutants (produced by the rule-based tool MutPy) to generate tests that kill them, halting when no survivors remain; it is single-agent, benchmark-scale, and reports mutation score with no dollar cost. Meta ACH [arXiv:2501.12862, FSE 2025] uses three agents (fault generation, equivalence detection, and test generation) with a single backbone model to produce few, concern-specific mutants and hardening tests at industrial scale, under an assured generate-and-test filter. LLMorpheus [arXiv:2404.09952] instead generates the mutants themselves with a model, producing diverse bug-like mutants beyond a standard operator set, and a recent empirical study [arXiv:2406.09843] compares such model-generated mutants against real bugs at scale. Our mutants are rule-based (mutmut), as in MuTAP, rather than model-generated as in LLMorpheus and AdverTest; this keeps the injected defect independent of any model under test.

### 2.3 Adversarial and dual-agent test-versus-mutant loops

The closest neighbor is AdverTest [arXiv:2602.08146], which runs a genuine dual-agent adversarial loop: a test agent and a mutant agent iteratively attack each other under an execution-based mutation-kill oracle, with measured fault-detection gains over prior baselines. Crucially, both of AdverTest's agents share a single backbone model per configuration, and no cross-lineage configuration is measured. Our loop replicates this adversarial-under-a-mechanical-oracle pattern, and we claim it as a replication, not an invention (Section 5.1). What AdverTest and the survivor-feedback lineage of MuTAP do not do, and what our H2 compares, is a cross-provider critic configuration against a same-provider control under the mechanical oracle (a comparison

of the two named configurations; Section 3.2).

### 2.4 Cross-model diversity, self-preference, and evaluation bias

The premise that a mechanical oracle is worth its cost rests on known failures of model-graded evaluation: LLM-as-judge protocols carry position, verbosity, and self-enhancement biases [Zheng et al., arXiv:2306.05685, NeurIPS 2023], and large language model evaluators can recognize and favor their own generations [Panickssery, Bowman, and Feng, arXiv:2404.13076, NeurIPS 2024], a self-preference bias a model-graded verdict cannot escape. Heterogeneous-model methods such as multi-agent debate [Du et al., arXiv:2305.14325, ICML 2024] motivate the intuition that mixing model lineages helps, but the direction is genuinely open: Self-MoA reports same-model repeated sampling beating heterogeneous mixtures on reasoning benchmarks, and Refute-or-Promote [arXiv:2604.19049] argues a cross-model critic catches correlated blind spots but does so in defect discovery, without a controlled same-lineage ablation and without a purely mechanical oracle. Most directly relevant to our Discussion, recent work argues that models increasingly share the same errors, which undermines the reliability of using one model to oversee another [Great Models Think Alike and this Undermines AI Oversight, arXiv:2502.04313]. Our instrument autopsy (Section 5) is a concrete, receipted instance of a related but distinct hazard: even with model judgment removed from the verdict entirely, an asymmetry in the harness that feeds the two models can manufacture a large false cross-model difference.

### 2.5 Cost accounting and pre-registration

Where cost is reported in this literature, it is normalized to an input unit rather than an outcome. AdverTest reports an average dollar cost per method-generation run (for example $0.270 on Defects4J), and TestForge [arXiv:2503.14713] reports dollars per file and per iteration, but neither divides cost by mutants killed or faults detected. The nearest per-outcome figure is in an adjacent domain, defect discovery, where a cross-model critic system reports a per-discovered-vulnerability cost. We report cost per mutation kill, a normalization not found in the studies reviewed here. Separately, pre-registered and registered-report protocols have precedent in software engineering [arXiv:2302.03649] and have begun to wrap LLM-for-software-engineering studies [arXiv:2606.10702], though they remain uncommon in this literature; we adopt pre-registration as a matter of rigor and claim no priority for it.

### 2.6 Positioning

Among the studies reviewed here, none measures the effect of critic-generator lineage or provider diversity on test-suite fault detection under a mechanical mutation-kill oracle, and none reports LLM test-generation cost normalized per killed mutant. We do not claim to be first to run an adversarial test-versus-mutant loop under a mechanical oracle (AdverTest does this), first to feed survivor mutants into model prompts (MuTAP does this), or first to pre-register a software-engineering protocol. We claim a paired cross-provider versus same-provider critic-configuration comparison under a mechanical oracle (a configuration comparison, never an isolated lineage effect), a per-outcome cost normalization not found in the studies reviewed here, a causal within-replicate estimate of the Critic continuation's incremental effect together with a descriptive pipeline-level comparison in the MuTAP-to-AdverTest direction, and, as the durable contribution, a fully receipted autopsy of a provider-correlated instrument artifact. The broader literatures on

experimental-infrastructure validation and execution-environment nondeterminism are engaged here only where they touch this study's own defects; we do not survey them.

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## 3. Method

### 3.1 The adversarial test-hardening loop

We evaluate a two-role loop over a mechanical mutation-kill oracle. A Tester model is prompted once to write a pytest suite for a target module (round 0). The suite is run under mutation testing (mutmut): each surviving mutant is an injected defect the suite failed to catch. A Critic model is then handed the named survivors, the specific mutants left alive together with their diffs, and prompted to write tests that kill exactly those. The loop repeats. After each Critic round the suite is re-measured, newly killed survivors are removed from the target list, and the next round attacks what remains. A run stops when two consecutive rounds kill nothing new (dry_rounds = 2) or after four rounds (max_rounds = 4), whichever comes first.

The verdict at the core of this loop is mechanical: a generated test either kills a mutant under pytest or it does not, and no model ever judges its own or another model's output. This closes the self-preference channel that undermines LLM-as-judge evaluations, where a model scoring model output has no ground truth. We are careful not to overstate what this buys. The mechanical oracle removes model judgment from the verdict itself; it does not make the surrounding measurement apparatus neutral. Section 5 documents a case where the harness around the oracle opened a lineage-correlated channel into the result even though the oracle itself never rendered a subjective judgment.

We do not claim the loop is novel. Feeding survivor mutants into an LLM prompt was established by MuTAP, and a dual-agent adversarial test-versus-mutant loop under a mechanical oracle was established by AdverTest. Our contribution is a controlled measurement, described below, and a fully receipted account of what it took to make that measurement trustworthy (Section 5).

### 3.2 Arms and hypotheses

Each subject module is run through three arms, all sharing one fixed Tester model (claude-sonnet-5, Anthropic) so that the only deliberate variable is the Critic:

- oneshot: Tester round 0 only, no Critic round. This is the one-shot baseline.
- loop-same: Tester round 0, then Critic rounds where the Critic is the same lineage as the Tester (claude-sonnet-5, Anthropic).
- loop-cross: Tester round 0, then Critic rounds where the Critic is a different lineage (gpt-5.6-terra, OpenAI).

The cross-lineage Critic is deliberately price-matched to the same-lineage Critic (posted token rates \$2.50/\$15 per million versus \$3/\$15) rather than paired with the flagship variant, so that the comparison is not additionally confounded with price tier; we have no independent evidence of capability-tier equivalence across providers and claim none. Price matching cannot isolate lineage: the arms compare two named model-provider-harness configurations, and "cross-lineage" is used throughout as the descriptive label of the arms, never as the inferred cause of a difference. We pin real model identifiers throughout, rather than anonymizing, because the cost accounting

is tied to specific rate cards and the instrument finding in Section 5 is provider-specific and cannot be stated without them.

Two hypotheses follow:

- H1 (replication): the adversarial loop (loop-same) kills more mutants than one-shot generation (oneshot), holding the Tester model fixed.
- H2 (pre-registered wording): a cross-lineage Critic (loop-cross) kills more of the survivors the same-lineage Tester missed than a same-lineage Critic (loop-same) does. "Cross-lineage" is the pre-registered arm label; what the arms instantiate are the two named model-provider-harness configurations above, and every H2 statement in this paper is a claim about those configurations.

H1 is framed strictly as a replication of the MuTAP-to-AdverTest loop direction in a new agentic, repo-level, Python setting with disclosed per-outcome costs. H2 is the claim we could not find measured in the studies reviewed here: none measures the effect of critic-configuration diversity (cross-provider versus same-provider) on test-suite fault detection under a mechanical mutation-kill oracle. Both directions of the H2 result were pre-declared publishable, including a null, so the direction was not assumed in our favor.

One design limitation is inherent to these arms and we state it where the arms are defined rather than burying it later. Each arm independently regenerates its own round 0, so the tester suite and its resulting survivor set are not held fixed between loop-same and loop-cross. The two critic arms therefore differ in both the round-0 tester draw and the critic lineage, and single-run tester draws carry substantial variance (Section 6 reports the same tester producing round-0 kill counts of 69, 168, 50, 62, and 65 across repeated attempts on one subject). A cleaner design freezes one round-0 survivor set and runs every continuation from that identical state. Experiment 1 did not, so its H1 and H2 estimates are comparisons of independently sampled pipelines, and any small cross-lineage difference it reports must be read against that confound. An external cross-model review of this manuscript subsequently identified exactly this as blocking a causal reading, and Experiment 2 (Section 3.10) is the frozen-round-0 design that removes it.

### 3.3 Subjects and selection

Five Python subjects, each pinned by commit SHA, each contributing one paired subject-module cell to every arm. Subjects were selected against criteria fixed before selection (recorded in the pre-registration's selection log). The two third-party subjects were drawn by scanning the most-downloaded PyPI packages in rank order and taking the first two that met every pre-registered criterion with no discretion: pure Python (no C extensions), a permissive license, an existing pytest suite (which we strip in the clone), a plain-logic module of 100 to 800 source lines (no network- or IO-heavy modules; the first qualifying module in alphabetical order becomes the target), at least 40 mutants generated by mutmut on that module, and no prior authorship, contribution, or analysis by the author. The three author-repository subjects are a convenience sample chosen for thesis relevance, including the degenerate zero-kill baseline case described below. This convenience component is a disclosed threat to external validity (Section 6), compounded by training-data contamination since all five are public code the models may have seen in pretraining. For each subject exactly one target module is mutated, and sibling files needed for imports are carried into the mutation sandbox unmutated. One subject (graph-guard) was pre-declared as a pilot with a go/no-go gate before the full grid, and its pilot cells were pre-declared to count as

data; in the event those cells were later reclassified for an instrument defect (Section 3.8) and rerun, so the pilot designation does not affect the counted dataset.

| Subject | Provenance; suite | Target module | Mutants | Killed by existing suite | Baseline survivors (scored) |
|---|---|---|---|---|---|
| attrition-risk-ml | author repo (public); kept | `src/train.py` | 255 | 0 | 255 |
| graph-guard | author repo (public); kept | `graph_guard/ppr.py` | 80 | 58 | 22 |
| rag-guard | author repo (public); kept | `rag_guard/guard.py` | 71 | 46 | 25 |
| packaging | pypa/packaging (OSS); stripped | `src/packaging/_elffile.py` | 69 | 0 | 69 |
| idna | kjd/idna (OSS); stripped | `idna/cli.py` | 187 | 0 | 187 |

Both mutation denominators are shown, per the pre-registration's rule that neither the full-mutant count nor the baseline-survivor count is presented alone. The two third-party subjects have their existing suites stripped in the local clone before any run, so the loop is scored against a genuinely empty starting suite and never against the upstream project's own tests; the three author-repo subjects keep their existing suites, so their scored denominator is the survivors an already-present suite left alive.

One subject, attrition-risk-ml, is degenerate by design and kept deliberately. Its existing suite kills 0 of 255 mutants, because the target module is never imported by any test, so it is the maximal-headroom false-pass case: a suite that looks present but catches nothing, exactly the failure mode this line of work exists to expose. Because any ratio against a zero-kill baseline is undefined and any intervention trivially wins against zero, this subject is reported with absolute kill counts only, never a relative-improvement figure.

We perform no equivalent-mutant detection. Some mutants in each subject may be semantically equivalent to the original and therefore unkillable by any test, which means every reported kill rate is a lower bound. This does not affect the paired comparisons: both arms in any comparison face the identical mutant set, so an unkillable mutant simply lands in the both-miss cell that McNemar discards.

### 3.4 The mechanical oracle

Mutation testing is run with mutmut, which injects rule-based, single-change mutants (one mutation per variant), in contrast to LLM-generated mutants used by some adversarial-loop systems. Each subject's mutation scope, meaning the exact module mutated and the exact sibling files copied into the sandbox, was fixed at pre-registration and thereafter changed only through logged, receipted amendments (Section 3.8), never silently at run time. Kills are counted per mutant over each subject's own pristine-baseline survivor set: the survivors measured on

the pristine module before any generated test exists, so both arms in any comparison are scored against the same universe of mutants. A generated test file is rejected outright, crediting zero kills with the rejection and its reason recorded, when it fails validation as a file: it does not parse, the reply was truncated or malformed, or none of its tests survive the per-test checks. Within an accepted file, each individual test must pass on the pristine (un-mutated) module twice (the flake check); a test that fails on pristine code, a wrong-oracle test rather than a bad file, is pruned individually and logged, never counted as a kill, and the round continues with the tests that survive (a file that passes the pristine run once and fails the repeat is rejected whole as flaky – a code path no counted round ever triggered).

### 3.5 The paired statistic

Because every arm runs against the same pinned subject clone, kill outcomes are paired mutant-for-mutant across arms. The primary statistic is a two-sided exact McNemar test on the discordant pairs: b = mutants killed by arm A only, c = mutants killed by arm B only. Pairs where both arms agree (both kill, or both miss) carry no information and are excluded by construction; the evidence lives entirely in the disagreements. The test uses the doubled exact binomial tail (min-tail doubling, with $n = 0$ defined as $p = 1$), not the chi-square approximation, which is appropriate for the small discordant counts these subjects produce.

The pooled test treats individual mutants as the unit of analysis, which assumes independence across mutants; in fact mutants are clustered by subject (idna alone contributes 64 of the 105 H1 discordant pairs), and within-subject correlation among mutants of one module is not modeled. This inflates the nominal pooled significance, which is why we pre-registered per-subject tables as a companion readout and report them alongside every pooled figure rather than resting on the pool alone.

Each hypothesis is reported in three pre-declared views, all fixed before any data. First, one pooled McNemar across all subjects' discordant pairs, designated the confirmatory statistic. Second, a pooled view excluding the degenerate subject, whose 255 paired outcomes would otherwise dominate the pool. Third, per-subject McNemar tables. All three views were committed in advance and all three are reported for both hypotheses, so no view can be chosen, promoted, or dropped after seeing results. We ran no a priori power analysis; with three of four scorable subjects at or near a 98.6 to 100 percent kill ceiling, the number of discordant pairs available to H2 was near its structural minimum, so the H2 null is stated throughout as "no effect detectable at this design's power," never as "no effect."

### 3.6 Pre-registration, enforced by tooling

The protocol, meaning the hypotheses, arms, subjects, frozen scopes, the exact statistic, and the supported-or-not-supported criteria, was written and committed before any paid run. The freeze is not a promise. The experiment command mechanically refuses to run any arm unless the committed machine-readable protocol is byte-identical to the repository HEAD, and every reported number is required to be recomputable from committed per-run receipts (the per-round JSON logs recording token counts, cost, verdicts, and preserved test artifacts for every run). The success criteria were fixed in advance: H1 is supported if and only if the pooled McNemar yields $p < 0.05$ and the direction favors loop-same; H2 if and only if the pooled McNemar yields $p < 0.05$ and the direction favors loop-cross. Any other outcome, in either case, is not supported,

and a null was pre-declared to be reported with the same prominence as a positive result. No subgroup analysis was permitted to substitute for the three pre-declared views; any post-hoc breakdown is labeled exploratory and never replaces the primary readout.

Each arm runs once per subject under a fixed grid (three arms times five subjects, fifteen cells). A cell that produces a valid verdict is never rerun in search of a better one, and there is no early stopping for significance. A cell that is invalidated or aborts may be rerun only under a logged deviation. One subject's loop-same cell was attempted five times; before the final attempt, a stopping rule was pre-committed in the deviations log (if that attempt also failed, the cell would be scored missing, with no further gates), and when it failed the cell was recorded as missing data rather than backfilled or estimated. Every departure from the frozen protocol is logged in an append-only deviations file. Fourteen of the fifteen cells produced valid verdicts; the one missing cell (attrition-risk-ml loop-same) removes the largest subject from the paired statistics, so all paired analyses span four subjects, and the pre-declared with-degenerate and without-degenerate pooled views coincide in composition for this dataset (a composition collapse we state explicitly).

### 3.7 Cost accounting

Every model round is metered exactly from its input and output token counts; an unpriced model fails the run closed rather than pricing at a guessed rate. We report cost-per-kill, meaning total cost divided by mutants killed, per arm. This per-outcome normalization is not found in the studies reviewed here, which report cost per method-run or per file; the adjacent defect-discovery literature does report a per-outcome figure (a cross-model critic system reports a dollar cost per discovered vulnerability), so we scope the novelty to test generation rather than claiming it broadly.

### 3.8 Instrument integrity

The measurement instrument was itself treated as a subject of scrutiny, and the honest account of what that surfaced falls into three tiers rather than one.

First, defects caught at zero model cost, before any model was called, by a model-free dry-run validator built partway through the study: scope-transcription errors and a mutation-sandbox failure that would have silently recorded every mutant as unchecked. From the point that validator existed, and later a canary must-kill probe that confirms the sandbox actually collects a freshly written test, each subsequent scope fix was validated by a zero-cost dry run before any further paid cell ran.

Second, defects caught only after paid cells had already run, which forced reclassification of already-spent data. A trampoline crash that laundered into a false all-survived zero, and an include-list scoping bug that silently prevented freshly generated tests from being collected, together invalidated several paid cells, including two that had previously been counted as data. More was ultimately spent on runs that were invalidated and rerun than on the runs finally counted; we report the invalidated spend alongside the counted spend rather than omitting it.

Third, the one defect caught only after all fifteen counted cells had run, by a final adversarial review of the completed analysis rather than by any runtime gate. Because it changed the headline result, it is documented in full as a finding in its own right (Section 5). In every tier the repair principle was the same: make the instrument fail loudly rather than record a plausible-

looking wrong number. Across the study the protocol carried ten numbered amendments plus one prompt-template amendment, each logged with its rationale and its cost consequences.

### 3.9 Reproducibility and disclosure

All counted cells ran under a single fixed prompt version, with one per-subject import hint added for the one subject whose package layout required it; the prompt texts are content-hashed and the hashes are recorded in the pre-registration (16-character prefixes: tester 98abbb8532990865, critic 703335c0af111836). The models were called at provider default sampling: no temperature or top-p value is set anywhere in the experiment's provider configuration, and the only sampling-relevant request-body parameter is the output-token cap discussed in Section 5. Mutation testing used mutmut 3.6.0 (pinned) with pytest 8 or newer; the package requires Python 3.11 or newer and its continuous integration validates on Python 3.11, 3.12, and 3.13. All runs executed on macOS on Apple Silicon, the platform whose fork behavior produced the missing-cell deadlock analyzed in Section 6. The protocol, the machine-readable arms configuration, the per-run receipts, the deviations log, and the analysis script that recomputes the reported statistics are committed to the public repository, so every reported statistic is reproducible from the receipts. A small, bounded portion of the all-in spend total is disclosed as an estimate rather than receipt-derived, where a crash billed a call before its receipt row landed. An automated reproduction command is provided. Experiment 2 adds its own artifacts under the same discipline: its protocol was committed and pushed to the public repository before its runner existed and before any paid run (the pushed commit is the pre-registration timestamp; we note this is a public, hash-chained record rather than a third-party registration service), the runner was required, as a pre-registered precondition, to enforce the freeze mechanically before the first paid call, every frozen seed is committed before its continuations may run (twenty of the twenty-one seeds were also pushed publicly before any of their continuations ran; the one mid-run replacement seed was committed locally before its continuations and pushed with the final receipts, so commit-before-continuation is the mechanically enforced gate and the public push is the audit trail), and the analysis script's bootstrap draw count (10,000) and random seed are frozen in the protocol file itself, so every reported interval is bit-for-bit reproducible. The manuscript was drafted by an AI assistant from these frozen artifacts under the author's direction, and every quantitative and design claim was checked against the committed sources; this tooling is disclosed rather than hidden, consistent with the paper's subject. The pre-registration's limitations (training-data contamination, no syscall sandbox against mutant-environment gaming, a two-run flake check that a one-third-flaky test still clears roughly four times in nine, a single fixed Tester model, one subject's stdin code path, and one subject's pinned-module divergence from its development branch) are carried into a dedicated Threats to Validity section rather than asserted away here.

### 3.10 Experiment 2: the frozen-shared-round-0 causal design

Experiment 2 removes the initial-draw confound with one mechanical change: the round-0 suite becomes frozen shared state instead of a per-arm sample. For each replicate, the Tester is called once; the resulting suite is validated by the standard guardrails, measured once, and frozen, the test file bytes, the pristine baseline survivor set, and the post-round-0 survivor set committed to the repository before any continuation may run (the same committed-byte-identity gate that enforces the protocol freeze is applied to each seed directory). Three continuations then run from that identical state: no-critic (the frozen round 0 is the outcome, at zero marginal cost),

loop-same (same-lineage Critic rounds), and loop-cross (cross-lineage Critic rounds), with the loop parameters of Experiment 1. The loop's incremental effect is therefore a within-replicate paired quantity: every continuation of a replicate starts from byte-identical tests and an identical survivor set, so arm differences cannot come from initial-draw variance. Five replicates per subject, drawn independently, average over the round-0 draw distribution itself.

Four of Experiment 1's five subjects are used, under the same pinned commits and frozen scopes; the degenerate zero-kill-baseline subject is excluded, pre-declared, both for its unresolved nested-parallelism deadlock (four failed attempts in Experiment 1) and because its special-case analysis rules add nothing to a causal estimate. The protocol pre-registers outcome-blind procedure for every contingency: seed draws are capped at three per replicate and seven per subject, counted mechanically from receipts on disk; a draw is accepted or rejected only by the mechanical validity gates, never by its kill count; a seeded continuation re-verifies the frozen measurement before any Critic call (the pristine baseline and the post-injection survivor set must reproduce exactly), and any mismatch invalidates the whole replicate, retires the seed in place, and makes a replacement draw mandatory if the caps permit; a validly scored cell is never rerun; an aborted or crashed cell is rerun exactly once, then scored missing. Every one of these rules is enforced by the tooling, not by the operator, and every departure is logged in the same append-only deviations file as Experiment 1.

The pre-registered estimands are effect sizes with uncertainty intervals, not p-value gates, and the causal quantity is deliberately narrow: the incremental mutation kills produced by this guarded, add-only Critic continuation, conditional on these frozen seeds, subjects, models, and stopping rules. Nothing broader, no cross-repository or cross-model generalization, is identified by the design. Because generated tests are add-only and both loop continuations start from the identical frozen suite, a continuation's incremental kills over the seed (Delta) cannot be negative by construction, so the pre-registered question is magnitude, never sign. We report, per replicate, Delta for each loop arm and the incremental kill rate rho = Delta / |S| over the frozen post-round-0 survivor set S (undefined when |S| = 0, an exclusion fixed in advance), and for the cross-lineage question the within-replicate differences D = Delta_cross - Delta_same and rho_gap = rho_cross - rho_same. Per subject, each estimand is summarized by its mean over the five replicates with a 95% percentile bootstrap interval; the pooled figure is the unweighted mean of the four subject means with a subject-level cluster bootstrap, resampling subjects, because replicates within a subject are correlated and mutants within a replicate far more so. No pooled per-mutant McNemar is used as a confirmatory statistic in Experiment 2, for exactly the clustering reason Section 3.5 discloses about Experiment 1's pooled view. The bootstrap draw count and random seed are frozen in the protocol file. The cross-lineage estimand is pre-registered as a pilot: no supported-or-not verdict attaches to it, because four subjects cannot power a lineage claim and because "lineage" here bundles model, provider API behavior, and harness interaction, a bundling the autopsy makes vivid.

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## 4. Results

Sections 4.1 through 4.5 report Experiment 1; Sections 4.6 through 4.9 report Experiment 2. Every figure below is recomputed from committed per-run receipts by the committed analysis scripts, driven for Experiment 1 by a committed manifest of which runs count and for Experiment

2 by the frozen seeds and receipts themselves. This analysis supersedes an earlier one: a final adversarial review of the completed analysis found a truncation defect sitting inside the counted cells (an output cap with no truncation detection), which reclassified four counted cells as instrument-invalid and confounded the entire H2 readout. The version reported here is computed from the corrected instrument with the affected cells rerun. The headline difference is itself the paper's central finding: H2's previously significant result did not survive the correction, and the Discussion analyzes why.

### 4.1 Counted cells

The design is five subjects times three arms, fifteen cells. Fourteen produced valid verdicts; one, attrition-risk-ml loop-same, is a missing cell, after its original run was reclassified instrument-invalid and four rerun attempts each failed on a nested-parallelism deadlock that survived three targeted, individually verified instrument fixes; the missing verdict was recorded under a stopping rule pre-committed before the final attempt. Kill counts are per mutant over each subject's own pristine-baseline survivor set. The verdict column records how each run terminated: oneshot (baseline, no critic round); clean (all baseline survivors killed); dry (two consecutive rounds killed nothing new); rounds-cap (reached the four-round limit); missing (no valid verdict within the attempt budget). Dropped counts wrong-oracle tests pruned across all rounds and never credited as kills.

| Subject | Arm | Verdict | Baseline survivors | Killed | Rate | Cost ($) | Dropped |
|---|---|---|---|---|---|---|---|
| graph-guard | oneshot | oneshot | 22 | 12 | 54.5% | 0.13 | 0 |
| graph-guard | loop-same | dry | 22 | 17 | 77.3% | 1.02 | 1 |
| graph-guard | loop-cross | clean | 22 | 22 | 100.0% | 0.35 | 3 |
| rag-guard | oneshot | oneshot | 25 | 21 | 84.0% | 0.06 | 0 |
| rag-guard | loop-same | clean | 25 | 25 | 100.0% | 0.11 | 0 |
| rag-guard | loop-cross | clean | 25 | 25 | 100.0% | 0.16 | 1 |
| packaging | oneshot | oneshot | 69 | 36 | 52.2% | 0.11 | 0 |
| packaging | loop-same | rounds-cap | 69 | 68 | 98.6% | 0.64 | 0 |
| packaging | loop-cross | dry | 69 | 68 | 98.6% | 0.30 | 3 |
| idna | oneshot | oneshot | 187 | 123 | 65.8% | 0.12 | 0 |
| idna | loop-same | clean | 187 | 187 | 100.0% | 0.65 | 9 |
| idna | loop-cross | clean | 187 | 187 | 100.0% | 0.19 | 0 |
| attrition-risk-ml | oneshot | oneshot | 255 | 62 | 24.3% | 0.07 | 0 |

| Subject | Arm | Verdict | Baseline survivors | Killed | Rate | Cost ($) | Dropped |
|---|---|---|---|---|---|---|---|
| attrition-risk-ml | loop-same | missing | 255 | n/a | n/a | 0.07 | 0 |
| attrition-risk-ml | loop-cross | rounds-cap | 255 | 254 | 99.6% | 0.34 | 2 |

Costs are shown to two decimals for readability; the aggregate figures in Section 4.5 are computed from the full-precision receipts, so summing the rounded column will not exactly reproduce them. Nineteen wrong-oracle tests were pruned across the fourteen valid cells (nine of them in idna loop-same alone), each logged and never counted as a kill.

Because attrition-risk-ml's loop-same cell is missing, that subject contributes no paired data to either hypothesis, so all paired statistics below span the remaining four subjects. The pre-declared pooled-with-degenerate and pooled-without-degenerate views therefore coincide in composition for this dataset, a composition collapse we state explicitly.

### 4.2 H1 (replication): does the loop kill more than one-shot?

Pre-written support criterion (fixed before any run): H1 is supported if and only if the pooled McNemar test comparing loop-same to oneshot yields $p < 0.05$ and the direction favors loop-same ($b > c$, where b counts mutants killed by loop-same only).

Per-subject paired 2x2, over each subject's baseline survivor set:

| Subject | both | b (loop-same only) | c (oneshot only) | neither | p (exact McNemar) |
|---|---|---|---|---|---|
| graph-guard | 12 | 5 | 0 | 5 | 0.0625 |
| rag-guard | 21 | 4 | 0 | 0 | 0.125 |
| packaging | 36 | 32 | 0 | 1 | 4.66e-10 |
| idna | 123 | 64 | 0 | 0 | 1.08e-19 |

Pooled (confirmatory view): $b = 105$, $c = 0$, $p = 4.93e\text{-}32$.

Under the pre-registered criterion this reads as supported, and we report that for fidelity to the pre-registration; we then immediately bound it. The pooled unit of analysis is the mutant, mutants cluster within four subjects, and the arms are independently sampled pipelines (Section 3.2), so the pooled inference is not valid for generalization beyond these subjects and this comparison is descriptive: the loop-arm pipeline outscored the one-shot pipeline on every subject, and across 303 paired baseline survivors not one mutant was killed by one-shot generation but missed by the loop arm. The causal estimate of the loop's incremental effect is Experiment 2's (Section 4.7), not this table's. Two subjects individually clear $p < 0.05$ (packaging, idna) and two do not (graph-guard at 0.0625, rag-guard at 0.125, both with small discordant counts), but the direction is consistent in all four. The pooled test treats mutants as independent when they in fact cluster by subject, which inflates the nominal pooled significance; the pool is also

idna-heavy (64 of the 105 discordant kills) and packaging-heavy (32 of 105). We therefore report the per-subject tables alongside the pool rather than resting on the pooled figure.

Notably, the instrument correction strengthened H1. Under the defect-affected instrument the H1 split read b = 79, c = 5, with one subject tied and one point against the loop. The truncation defect had been silently removing critic rounds that loop-same should have run, suppressing the loop's own kills. Correcting it moved packaging from a 4-to-4 tie to plus 32 and graph-guard from 0-versus-1-against to plus 5 (one accepted critic round of over 30,000 output tokens, impossible under the old cap, killed 5 survivors). The corrected cells are new stochastic draws, so no monotone counterfactual relation to the truncated originals follows; we note only that the correction moved the comparison further in the loop arm's favor.

### 4.3 H2: does the cross-provider critic configuration kill more? (pre-registered as "cross-lineage")

Pre-written support criterion: H2 is supported if and only if the pooled McNemar comparing loop-cross to loop-same yields $p < 0.05$ and the direction favors loop-cross. A null was pre-declared to be reported with the same prominence as a positive result.

Per-subject paired 2x2:

| Subject | both | b (loop-cross only) | c (loop-same only) | neither | p (exact McNemar) |
|---|---|---|---|---|---|
| graph-guard | 17 | 5 | 0 | 0 | 0.0625 |
| rag-guard | 25 | 0 | 0 | 0 | 1.0 (no discordant pairs, ceiling) |
| packaging | 68 | 0 | 0 | 1 | 1.0 (no discordant pairs) |
| idna | 187 | 0 | 0 | 0 | 1.0 (no discordant pairs, ceiling) |

Pooled (confirmatory view): b = 5, c = 0, p = 0.0625.

Verdict: H2 NOT SUPPORTED. The pooled p of 0.0625 does not clear the pre-registered 0.05 bar. The point direction favors loop-cross (all five discordant pairs, from the single subject graph-guard, go the cross-lineage way, and none go the other way), but the evidence does not meet the pre-written criterion. This is an underpowered null, not a demonstration of no effect: three of the four scorable subjects saturated at 98.6 to 100 percent kill in both loop arms, leaving zero discordant pairs, so the design had almost no room to detect a cross-lineage difference even if one exists. The correct reading is no effect detectable at this design's power, with a directionally positive but non-significant residual on one subject.

### 4.4 The central finding of Experiment 1: H2's earlier signal was an instrument artifact

Reported at the same prominence as any supported result, because it is the paper's central contribution. The superseded analysis reported H2 SUPPORTED at a pooled p of 9.5e-66, with b = 217 and c = 0, an apparently overwhelming cross-lineage effect. The corrected instrument erased essentially all of it.

- packaging: b went from 32 to 0. The old loop-same cell's two critic rounds had both been silently truncated at exactly 16,000 output tokens each and rejected under unrelated-sounding notes, crediting zero kills. Rerun with truncation detection and a raised cap, the same-lineage critic's rounds were accepted and killed 32 more survivors, landing at 68 of 69, identical to loop-cross. The strongly supported reading is that the cap deleted the verbose model's rounds; because this rests on a single rerun of a cell class that carries real run-to-run variance, tester-round variance as a partial contributor cannot be fully excluded by the receipts, so this is the well-supported reading rather than a mechanical decomposition.
- attrition-risk-ml: b went from 185 to excluded. Its old loop-same cell was likewise truncation invalidated, and the subject could not produce a valid replacement within the attempt budget, so the missing-cell rule removed it from the comparison entirely.
- What remains is graph-guard's b = 5, a real gap between two accepted runs worth noting descriptively, and three subjects at or near ceiling with no discordant pairs at all.

In summary: the dramatic cross-lineage effect this experiment appeared to measure was manufactured by a lineage-correlated instrument failure and does not survive the instrument's correction. The output cap sat on every call to the same-provider model, so it truncated the verbose same-lineage critic and some tester rounds alike, never the terse cross-lineage critic; the bias it injected into H2 specifically ran through the same-lineage critic rounds, which is what manufactured the apparent cross-lineage gap. The Discussion analyzes the mechanism, the fail-loud repair, and why this generalizes to any cross-model comparison run through an asymmetric harness.

### 4.5 Cost per kill

Aggregated across counted cells with valid verdicts (total cost summed and divided by total kills summed per arm, from full-precision receipts, never averaged as a per-subject ratio):

| Arm | Total cost ($) | Total killed | Aggregate $/kill |
|---|---|---|---|
| oneshot | 0.4844 | 254 | 0.0019 |
| loop-same | 2.4170 | 297 | 0.0081 |
| loop-cross | 1.3293 | 556 | 0.0024 |

One asymmetry must be read alongside the 3.4-times gap between loop-cross and loop-same: the oneshot and loop-cross aggregates each span all five subjects, while loop-same spans only its four valid cells (the missing attrition-risk-ml cell). loop-cross's 556 kills therefore include attrition-risk-ml's 254, which loop-same never reached, so the arms are aggregated over different subject sets and the ratio is not a like-for-like per-subject comparison.

Because the arms aggregate over different subject sets, we draw no comparative conclusion from this table. Experiment 2's like-for-like accounting (Section 4.9) supersedes it: there, token receipts show that output length and rejected rounds account arithmetically for most of the billing difference between these two configurations, with no inference about what capability- or budget-matched models would cost.

### 4.6 Experiment 2: execution, and the confound quantified

All twenty replicates completed with all three continuations validly scored: twenty-one seeds were frozen (twenty plus one replacement, below), no seed draw was rejected, and no cell is missing. The design's premise is visible in the seeds themselves. Five independent round-0 draws per subject produced kill counts of 29, 32, 35, 36, and 45 (of 69 baseline survivors) on packaging, a sixteen-kill spread from sampling alone, against a flat 12 (of 22) on every counted graph-guard draw and 127 to 132 (of 187) on idna: the initial-draw variance that Experiment 1's arms resampled is large and strongly subject-dependent, which is precisely why its comparisons could not isolate the loop. Every subject's pristine baseline measured identically on all of its draws, and each seeded continuation re-verified its frozen measurement before any Critic call.

The pre-registered contingency machinery fired twice, and both events ran to completion without operator discretion. First, one continuation's reproduction check failed: a mutant the frozen seed had measured as surviving was measured as killed at injection time, a test non-deterministic at the margin that had cleared the two-run flake check (the disclosed residual class). The cell was receipted as invalid and never scored, the replicate was invalidated wholesale, the seed was retired in place, and the mandatory replacement draw and its three continuations ran under the caps. Second, one continuation was killed mid-run by a tooling timeout external to the instrument; it was receipted as a crashed attempt and consumed the single pre-registered rerun, which completed cleanly. Both events are logged in the deviations file with their receipts.

### 4.7 The causal estimate: the loop's incremental effect over a frozen round 0

Per subject, the mean incremental kills (Delta_same) and incremental kill rate (rho_same) of same-lineage Critic rounds over the frozen round 0, with 95% bootstrap intervals over five replicates each; pooled values are unweighted means of subject means with subject-cluster bootstrap intervals:

| Subject | mean Delta_same [95% bootstrap interval] | mean rho_same [95% bootstrap interval] |
|---|---:|---:|
| graph-guard | 5.0 [5.0, 5.0] | 0.50 [0.50, 0.50] |
| idna | 57.2 [55.0, 59.0] | 0.99 [0.97, 1.00] |
| packaging | 24.8 [11.8, 34.0] | 0.77 [0.39, 0.97] |
| rag-guard | 4.4 [3.2, 5.4] | 0.87 [0.77, 0.97] |
| **pooled** | **22.9 [4.7, 44.2]** | **0.783 [0.592, 0.935]** |

Starting from an identical frozen suite, same-lineage Critic rounds killed a mean 78% of the survivors that suite left standing, and the pooled interval's lower bound sits far from zero (under the stricter two-stage bootstrap of Section 4.8, rho_same's interval is [0.583, 0.964] and Delta_same's is [4.65, 44.55], both materially unchanged). Because Delta is non-negative by

construction, sign carries no information and the pre-registered question was magnitude; the magnitude is large on these subjects but heterogeneous: graph-guard's loop killed exactly 5 of its 10 remaining survivors in every replicate (a hard stop at its numeric-tolerance mutants), while idna's loop swept nearly everything. The estimand this identifies is Section 3.10's narrow one: the incremental kills of this guarded, add-only continuation, conditional on these frozen seeds, subjects, models, and stopping rules. Within that scope it is the causal counterpart of Experiment 1's H1, isolated from the initial draw.

### 4.8 The cross-configuration pilot: per-replicate values, two readings, and sensitivities

The within-replicate differences between the two critic configurations, pre-registered as a pilot with no verdict attached. The complete replicate-level values (this comparison's raw data):

| Subject | D per replicate (cross - same, kills) | rho_gap per replicate | subject mean D | subject mean rho_gap |
|---|---|---|---|---|
| graph-guard | 5, 5, 5, 5, 2 | .50, .50, .50, .50, .20 | 4.4 | 0.440 |
| idna | 0, 0, 2, 1, 0 | 0, 0, .036, .017, 0 | 0.6 | 0.011 |
| packaging | 0, 0, 0, 39, 0 | 0, 0, 0, .975, 0 | 7.8 | 0.195 |
| rag-guard | 0, 1, 0, 0, 1 | 0, .167, 0, 0, .167 | 0.4 | 0.067 |

Pooled (unweighted subject means, pre-registered cluster bootstrap): D = 3.3 [0.5, 6.1]; rho_gap = 0.178 [0.039, 0.347]. Both pooled intervals sit above zero. Under Experiment 1's independently-sampled design the same comparison returned p = 0.0625 on five discordant pairs.

This pilot estimand is operational by pre-registration: every receipted failure counts, so it estimates end-to-end deployability of the two configurations under the specified harness, caps, and validators, not critic capability. The replicate table shows why the distinction matters, because the pooled gap decomposes into two mechanisms the receipts separate. On graph-guard, the same-provider critic's rounds were mostly accepted (16 of its 18 critic rounds across the five cells; the two rejections were themselves 32,000-token truncations, in replicates 1 and 2) and the arm plateaued at exactly 5 kills in every replicate, including the three whose rounds were all accepted, against the cross-provider critic's near-sweeps: evidence consistent with a capability-related difference on that subject's numeric-tolerance mutants, though we do not report per-arm generated-test volumes or token budgets and therefore do not claim an identified capability effect. On packaging, the entire subject mean rides on one replicate: the same-provider critic's first round was rejected for a formatting violation and its second hit the corrected instrument's raised output cap at exactly 32,000 tokens (the third such truncation in Experiment 2, and the only one that zeroed a cell's incremental kills), where the truncation detector added by the autopsy's repair caught it, archived the truncated reply, credited zero kills, and ended the cell at Delta_same = 0, while the cross-provider critic, from the identical frozen state, killed 39 of the 40 remaining survivors. That single replicate contributes 39 of the 66 total gap kills (59%). The mechanism that manufactured Experiment 1's false $p = 9.5 \times 10^{-66}$, verbose replies meeting a

mechanical output ceiling, is therefore a recurring operational property of this model-harness pairing, and detection makes it visible data rather than a laundered artifact, but detection does not remove it from the operational estimate.

Because of that leverage, we report post-hoc sensitivities (labeled exploratory; the pre-registered analysis is the pooled figure above). Excluding the truncation replicate: D = 1.35, rho_gap = 0.129. Leave-one-subject-out (point estimates only; no intervals attached): D ranges 1.8 (dropping packaging) to 4.3 (dropping rag-guard); rho_gap ranges 0.091 (dropping graph-guard) to 0.234 (dropping idna). Under a stricter two-stage bootstrap that resamples subjects and replicates within subjects, rho_gap's interval is [0.017, 0.368] on the full data and [0.004, 0.345] excluding the truncation replicate; D's is [0.3, 9.0] and [0.1, 3.5]. The honest summary: the direction of the configuration difference is stable under every sensitivity we ran, never crossing zero, while its magnitude is dominated by a single receipted operational failure and its interval, under the stricter bootstrap, approaches zero. A capability-oriented reading should rest on the graph-guard pattern (accepted rounds present in every cell, a persistent kill gap) and treat the packaging replicate as deployability evidence; neither reading supports a lineage inference, because the arms bundle model, provider API behavior, and harness interaction by construction.

Operationally, the cross-provider arm also failed less overall: 13 of its 20 cells ended with every survivor killed against 5 of 20 for the same-provider arm, with 3 rejected rounds against 10.

### 4.9 Experiment 2 cost, like for like

Experiment 2's arms span identical subjects and replicates, so its cost comparison has none of Experiment 1's denominator asymmetry. From full-precision receipts, on explicit bases: the twenty accepted seeds entering the analysis cost $2.20 (all seed draws including the one retired seed: $2.33); the same-lineage arm $11.14 for 457 incremental kills ($0.0244 per incremental kill); the cross-lineage arm $2.01 for 523 incremental kills ($0.0038 per incremental kill), 6.4x cheaper per outcome at a similar rate card. The spend entering the analysis (accepted seeds plus validly scored continuations) is $15.34; the all-receipted total is $15.47; the difference is the retired seed draw ($0.13), the invalidated cell and the crashed attempt having billed nothing before they stopped. The receipts decompose the difference arithmetically: the same-lineage critic's rounds emitted 678,564 output tokens against the cross-lineage critic's 100,545 (6.7x), and at the two configurations' similar output rates that output alone accounts for $10.18 of the $11.14 same-lineage arm cost and $1.51 of the $2.01 cross-lineage arm cost; the same-lineage critic also averaged 3.3 Critic rounds per cell against 2.45, with ten rejected rounds against three. Output volume and failed rounds, not per-token price, separate these two configurations' costs; we make no claim about capability- or budget-matched models.

All spend in one table, on explicit bases (each figure is derived in the sections above):

| Basis | Experiment 1 | Experiment 2 |
|---|---:|---:|
| counted / analysis-entering | $4.23 | $15.34 |
| all-receipted (incl. invalidated and rerun work) | $11.05 | $15.47 |
| disclosed unreceipted estimate (crash-billed calls) | ~$0.63 | $0 |

## 5. Discussion

### 5.1 What the two experiments establish about the loop

Experiment 1's H1 is a pipeline-level replication: the adversarial loop, as a pipeline including its own initial draw, outscored one-shot generation 105 discordant mutants to zero, confirming in a new setting (agentic, repo-level, Python, with a mechanical mutation-kill oracle) the direction established by MuTAP and AdverTest. It is not a causal estimate of the loop, because each arm sampled its own round 0. Experiment 2 supplies the causal version: with the round-0 suite frozen and shared, the Critic loop's incremental effect is a within-replicate paired quantity, and it is large, a mean 78% of remaining survivors killed (pooled rho_same 0.783 [0.592, 0.935]), positive on nineteen of twenty replicates (the twentieth is the truncation cell of Section 4.8), and heterogeneous across subjects in an informative way (near-total on idna, a hard 50% stop on graph-guard's numeric-tolerance mutants). Replication in a new setting is a contribution but not a discovery; the causal isolation, the per-outcome costs, and the fact that the estimate survived an instrument-correction process and then a design-correction process are what this pair of experiments adds. The two experiments estimate related but distinct quantities under different designs and budgets; Experiment 2's is the one its design identifies, and it is the estimate this paper stands on.

### 5.2 The instrument autopsy

Section 4.4 reports the incident; this section interprets it. The defect's essential property was not merely its size but its correlation with the H2 arm. The ceiling applied to same-provider calls generally; for H2, its consequential failures were verbose same-lineage Critic rounds, so it preferentially deleted work from one arm. A treatment-independent failure would not preferentially depress that arm; this one could manufacture a difference – and the manufactured result was invisible from inside: a clean, well-formatted, enormous p-value, with nothing in the number announcing that it measured the harness rather than the models. When detection was added, the four affected cells were invalidated; corrected-instrument replacements were run where obtainable, with the one unresolvable cell scored missing, and the apparent effect collapsed from $p = 9.5 \times 10^{-66}$ to p = 0.0625 (Section 4.4 gives the per-subject decomposition and the variance caveat on the single-rerun cells).

**How it was caught.** Not by a runtime gate. This study caught three earlier silent-corruption mechanisms mechanically, each before any number reached a results file: a sandbox crash that would have laundered into a false all-survived zero, and a scoping bug that would have silently prevented freshly generated tests from ever being collected, among them. The truncation defect slipped all of those gates, because it produced no crash and no error, only a plausible rejection note. It was caught by a final adversarial review that read the completed analysis, after every counted cell had already run, and asked why exactly these rounds were failing. That a defect this consequential survived every mechanical check and was caught only by an independent skeptical read is itself part of the finding.

**The fail-loud repair.** The fix makes the instrument halt loudly rather than emit a plausible wrong number. The output ceiling is now a single mechanical value read by both the request and a new check: every reply's output-token count is compared against it, and a reply at the ceiling raises a truncation error that is never retried (retrying a capped call would bill again and likely truncate again). The round is recorded honestly as "truncated: output hit max_tokens cap,"

its real billed cost is metered, zero kills are credited, and the raw truncated reply is archived as preserved evidence. A truncation at a cell's tester round (round 0) fails that cell loudly as missing data rather than being silently recovered; a truncated Critic round is recorded as rejected with zero kills and the loop continues, which is exactly how Experiment 2's three recurrences were absorbed. The principle is fail-closed: when the instrument cannot measure, it must say so, not hand back a confident wrong number.

**A residual asymmetry we disclose rather than paper over.** The truncation check exists only where a mechanical ceiling is declared, which is the same-provider wrapper. The cross-lineage provider sends no such ceiling and is therefore never truncation-checked. If that provider ever truncated a reply against a server-side limit, this instrumentation would launder it exactly the way the original defect did. We have no evidence any such truncation occurred (every accepted cross-lineage round is far from any plausible ceiling), but the detection asymmetry is structural and we name it, because the entire point of this section is that undisclosed asymmetries are how false results get made.

### 5.3 The general lesson: cross-model comparisons can inherit the asymmetries of their harness

The mechanical oracle at the core of this design was adopted specifically to close the self-preference channel that undermines LLM-as-judge evaluation, where a model scoring model output has no ground truth. At the verdict level it did exactly that: no model ever judged another model's output, and the kill verdict is a seeded defect and a failing test. The autopsy's central point is that a different bias, one level out from the verdict, survived the very mechanism meant to make model-versus-model comparison safe. Closing the judgment channel did not make the comparison neutral, because the apparatus that fed the two models was not itself symmetric.

The recurrences in Experiment 2 (Section 4.8) sharpen the lesson: the identical mechanism, a verbose model meeting a mechanical output ceiling, fired three more times under the corrected instrument within days of the repair, including on packaging, the same subject whose truncated rounds had been Experiment 1's clearest illustration. Asymmetries of this class are standing properties of a model-harness pairing, not one-off bugs, and the difference between a false finding and an honest data point is whether the harness detects the event or launders it.

The lesson generalizes beyond the specific defect. A comparison is only as neutral as the apparatus that runs it, and an apparatus that treats the two models differently in any dimension correlated with the outcome can produce a difference that looks like a model difference and is not. Here the correlated dimension was output verbosity interacting with a truncation ceiling; in another study it could be a timeout, a token budget, a rate limit, a retry policy, or a parser that tolerates one model's formatting quirks and not the other's. The defect is dangerous precisely because it is invisible in the result: the manufactured p-value was clean, well-formatted, and enormous. Nothing about the number announced that it was measuring the harness rather than the models.

Two practices follow, and this study argues for both by having needed both. First, pre-registration of the hypotheses and success criteria, so that a result cannot be quietly reinterpreted after the fact and so that the difference between the pre-declared analysis and the corrected one is itself a visible, dated record. Second, and less commonly practiced, fail-closed instrumentation: a measurement pipeline should be built so that when a component cannot do its job, it halts loudly instead of emitting a plausible substitute. Most of the silent corruptions in this study

were caught because the instrument was progressively rebuilt to fail loud; the one that reached the counted data was the one place the instrument could still fail silently.

### 5.4 What changed between the experiments, and what can be attributed to it

Experiment 1's H2 and Experiment 2's cross-configuration pilot ask the same question of the same subjects with the same models, and return different answers: an inconclusive p = 0.0625 on five discordant pairs, then a rate gap of 0.178 with a bootstrap interval above zero. Three things changed together: the round-0 draw was frozen and shared (removing initial-draw variance whose scale the seeds measured directly, a sixteen-kill spread on packaging), replication went from one cell per subject-arm to five, and counted-analysis spend increased 3.6-fold ($15.34 for the cells entering Experiment 2's analysis against $4.23 for Experiment 1's counted cells; each experiment's additional invalidated or replaced spend is receipted separately). The design does not separate those contributions, so we do not attribute the changed answer to variance removal alone; we claim only that the paired, replicated design yields a more informative pilot estimate than the original, and that Experiment 1's p = 0.0625 was an inconclusive reading from a confounded design, not evidence of absence that Experiment 2 overturned.

The pilot framing still binds, for reasons fixed in the pre-registration and sharpened by the data. Four subjects cannot power a configuration claim, and the pooled interval, resampling four subject means, is honest but coarse; under the stricter two-stage bootstrap it approaches zero. The arms bundle model, provider API behavior, and harness interaction, and Section 4.8 shows the bundle concretely: over half the measured gap is one receipted operational failure of the same-provider configuration. And the gap concentrates on the two subjects with headroom, so subject selection materially shapes the pooled number. A confirmatory test needs more and harder subjects and, ideally, multiple models per provider; the frozen-round-0 design that would carry it now exists, is public, and is validated.

### 5.5 Why the autopsy and the review loop are the contribution

Publishing the $p = 9.5 \times 10^{-66}$ result would have violated no procedure: it was pre-registered, directionally as hypothesized, and produced by real runs. Publishing H1 as a causal finding would have violated none either; the first complete draft of this manuscript did, until a cross-model review of the draft caught the round-0 confound its authors and same-family reviewers had read past. The same practice prevented both: the instrument, the completed analysis, and the manuscript were each reviewed adversarially from outside the family that produced them, and each level caught a defect everything below it had missed – the manuscript-level catch generating Experiment 2 itself. Cross-model evaluation is now routine in software-engineering research and practice; this study documents, with receipts, concrete ways such evaluations silently go wrong, and what catching them costs and pays.

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## 6. Threats to Validity

### 6.1 Construct validity

The oracle measures mutant kills, which is a proxy for fault detection, not fault detection itself (the gap between mutation kills and real-fault detection is the classic caveat of the mutation

literature [Papadakis et al. 2019; Just et al., ISSTA 2014], and pass-on-pristine plus mutant-kill is a partial answer to the oracle problem [Barr et al. 2015], not a resolution of it). Two gaps follow. First, we perform no equivalent-mutant detection, so some counted survivors may be semantically equivalent to the original and unkillable by any test; every reported kill rate is therefore a lower bound. This does not distort the paired comparisons, because both arms in any comparison face the identical mutant set and an unkillable mutant falls in the both-miss cell that McNemar discards, but it does mean absolute kill rates should be read as floors. Second, we do not run generated tests inside a syscall sandbox, so a sufficiently adversarial generated test could in principle detect properties of the mutation harness itself (timing, file paths, environment markers) and pass or fail on that rather than on the mutant's behavior. Our guardrails (add-only tests, pass-on-pristine validation, a flake check, and wrong-oracle pruning) reduce this but do not mechanically close it. A third construct threat concerns the cost measure: cost-per-kill does not isolate lineage, because the same-lineage critic's cost is dominated by its output verbosity rather than its lineage, so the cross-arm cost gap reflects model output style as much as the critic mechanism.

### 6.2 Internal validity

Experiment 1's comparisons carry the confound we disclose where the arms are defined (Section 3.2): each arm independently regenerates its own round-0 tester suite, so its loop-same and loop-cross cells differ in both the tester draw and the critic lineage, and a single-run tester draw carries substantial variance (the same tester produced round-0 kill counts of 69, 168, 50, 62, and 65 across repeated attempts on one subject; Experiment 2's seeds later measured a 29-to-45 spread on packaging under identical prompts). Experiment 2 removes this confound for the questions it re-measures, and its own internal-validity events are disclosed in Section 4.6: one replicate was invalidated by the reproduction check when a marginally non-deterministic test that had cleared the two-run flake check measured differently at injection time, and was replaced under the pre-registered mandatory flow; one cell consumed its single permitted rerun after an external tooling timeout. Both flows were outcome-blind by construction. A residual construct note on the cross-lineage pilot: its measured gap includes the same-lineage critic's operational failures (truncation, formatting rejections), which we count deliberately as deployment-relevant but which a capability-only reading would need to separate.

The study's central internal-validity event is the instrument artifact of Sections 4.4 and 5.2. One residual remains: truncation is mechanically detected only for the provider that declares an output cap, so a server-side truncation on the other provider would still be laundered. We have no evidence any occurred (every accepted cross-lineage round is far from any plausible cap), but the detection asymmetry is structural. A further, smaller asymmetry: one subject's missing-cell rerun attempts ran with an added parallelism-control configuration that its other two arms did not, because that configuration was introduced to fight a deadlock; the kill oracle is a deterministic pass or fail with no channel from a parallelism backend to a verdict, but the instrument configuration across that subject's arms is not byte-identical, and we record it. Finally, all five subjects are public code that the models may have seen in pretraining; the mutant-kill metric blunts but does not eliminate contamination, since a memorized suite would still have to target the injected mutant to score a kill.

### 6.3 External validity

The strongest limits are on generalization. One cell, the same-lineage-loop arm of the largest subject (255 baseline survivors), failed to produce a valid verdict within the pre-committed attempt budget, and the fixed once-per-subject design has no backfill mechanism, so that subject is dropped from both hypotheses entirely and the pre-declared with-degenerate and without-degenerate pooled views collapse to a single four-subject composition; the miss is disclosed, not smoothed over. The remaining paired statistics therefore span four subjects, of which three are the author's own public repositories, a convenience sample; the study is single-language (Python) and holds a single tester model fixed across all arms, so it does not speak to whether a different tester lineage would change either result. Ceiling effects further bound what H2 could show: three of the four scorable subjects saturated at 98.6 to 100 percent kill in both loop arms, leaving no headroom to detect a cross-lineage difference. Two subject-specific notes also bear on generalization: one subject's target module reads standard input in its command-line path, which mutation and collection never exercise, and one subject's pinned module differs from its own development-branch version, so any future run against that branch would be a different subject, not a rerun.

### 6.4 Statistical conclusion validity

The pooled McNemar test treats individual mutants as independent units, but mutants cluster by subject (one subject supplies 64 of the 105 H1 discordant pairs), and within-subject correlation is not modeled; this inflates the nominal pooled significance, which is why per-subject tables are pre-registered and reported alongside every pooled figure. No a priori power analysis was performed, and the H2 result is an underpowered null: with so few discordant pairs available, $p = 0.0625$ is near the smallest two-sided value the design could produce, so the null means no effect detectable at this power, never no effect. Kill counts also carry real run-to-run variance from single-run cells, so per-subject point estimates should be read as noisy; for example, one subject's same-lineage-loop kills moved from 11 of 22 to 17 of 22 between its invalidated original run and its valid rerun. And the flake check accepts a test after it passes on pristine code twice; a test that is flaky roughly one third of the time still clears a two-run check roughly four times in nine (flakiness in real suites is pervasive and well-documented [Luo et al., FSE 2014]), so some residual flaky-kill noise is expected and is not separately modeled; Experiment 2's reproduction check caught exactly one such marginal test downstream, and its seed was replaced under the pre-registered flow.

Experiment 2's statistics carry their own bounds, stated in the protocol before its data existed. The pooled intervals cluster-bootstrap only four subject means; with four clusters, percentile coverage is unreliable and discrete, the pre-registered variant treats each subject mean as fixed rather than propagating within-subject sampling variation, and we therefore also report a post-hoc two-stage bootstrap (resampling subjects, then replicates within subjects) and leave-one-out sensitivities in Section 4.8, and use "bootstrap interval" rather than "confidence interval" throughout: these are descriptive uncertainty statements for this subject set, not population-level inference. The unweighted pooled mean is itself a descriptive average over two author projects and two third-party modules, not an estimate of any naturally defined population. Incremental kills are non-negative by construction (tests are add-only from a shared frozen suite), so sign carries no information and all conclusions are magnitude-with-interval statements. Round-0 draw variance proved strongly subject-dependent (large on packaging, nearly absent on graph-guard),

so the flat five replicates per subject buy unequal precision across subjects; the design fixed K in advance rather than adapting it, and the heterogeneity is reported rather than modeled.

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## Code and Data Availability

The tool, both pre-registered protocols, all per-run receipts and frozen seeds, the deviations log, and the analysis scripts are in the public repository https://github.com/Jott2121/crucible (MIT license). The reported statistics are recomputed from the committed receipts by `python scripts/check_receipts.py`.

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